SIT233 Final Project Report

Nirosh Ravindran

[1]Department of Cybersecurity

BSCP|CS|63|134
rnirosh134@cicracampus.net

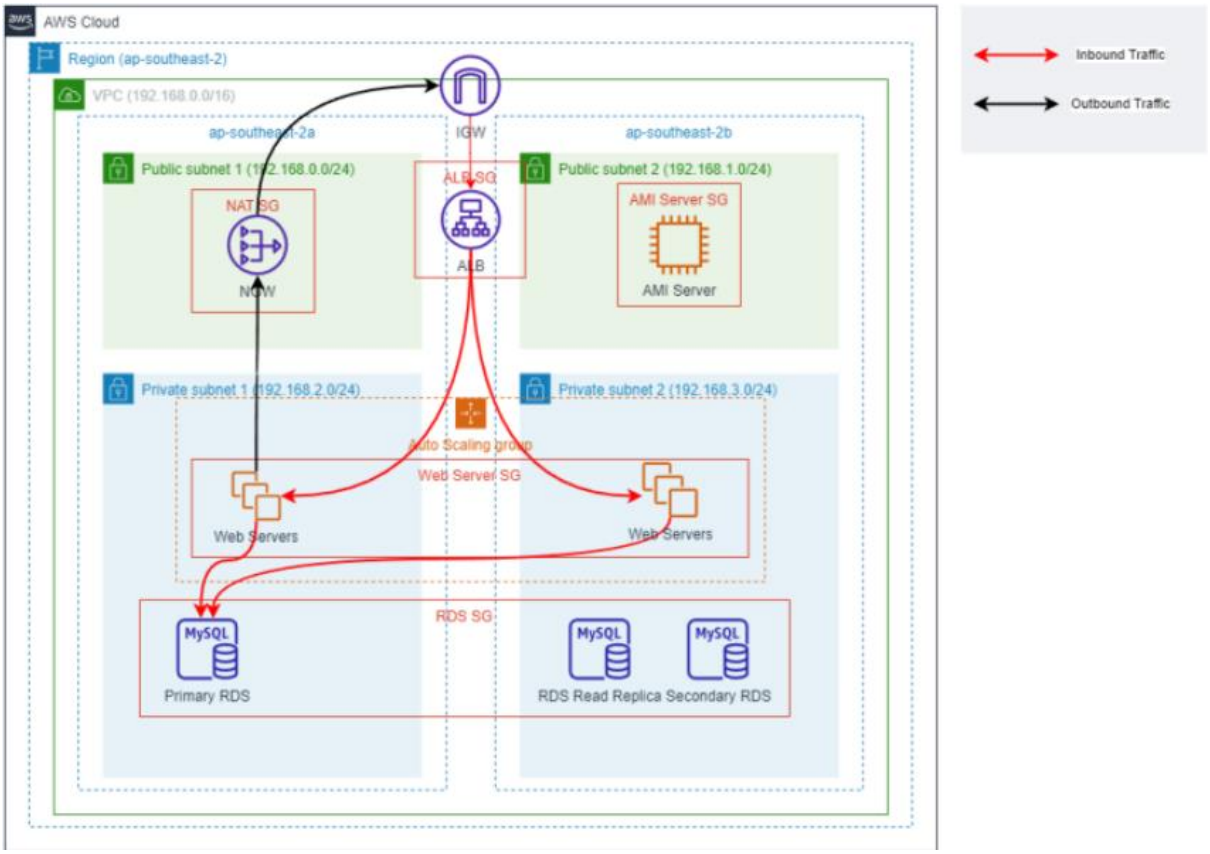
***Abstract*— This document offers a thorough account of the procedures followed in order to set up an AWS infrastructure for WordPress website hosting. A VPC, subnets, an internet gateway, and route tables are all created during the procedure. However, the project was unable to proceed past step 7 due to a problem with the formation of an IAM role. This paper describes the problems that were faced and the steps that were successful.**

***Keywords*— AWS, VPC, Subnets, Internet Gateway, Route Tables, IAM Role, WordPress Hosting**

I. INTRODUCTION

The method of setting up an AWS infrastructure to host a WordPress website is reported in this document. A Virtual Private Cloud (VPC), subnet configuration, internet gateway setup, and route table configuration are all part of the setup process. Sadly, a problem that arose during the creation of an IAM position forced the project to be suspended and prevented it from being completed.

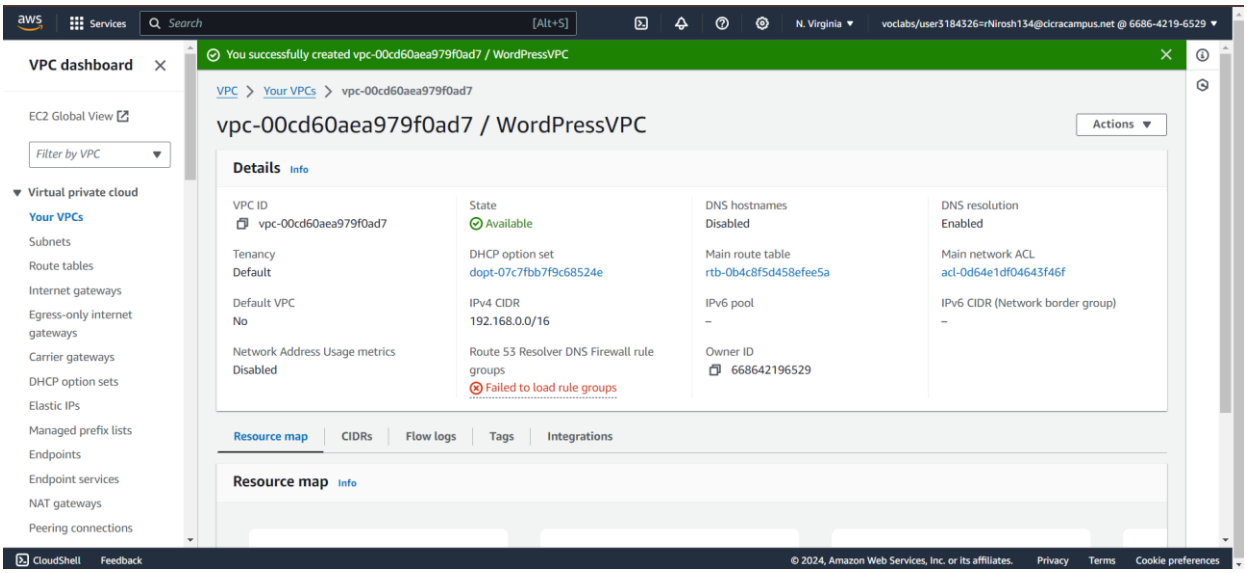
II. DESIGN DIAGRAM



III. **AWS INFRASTRUCTURE SETUP**

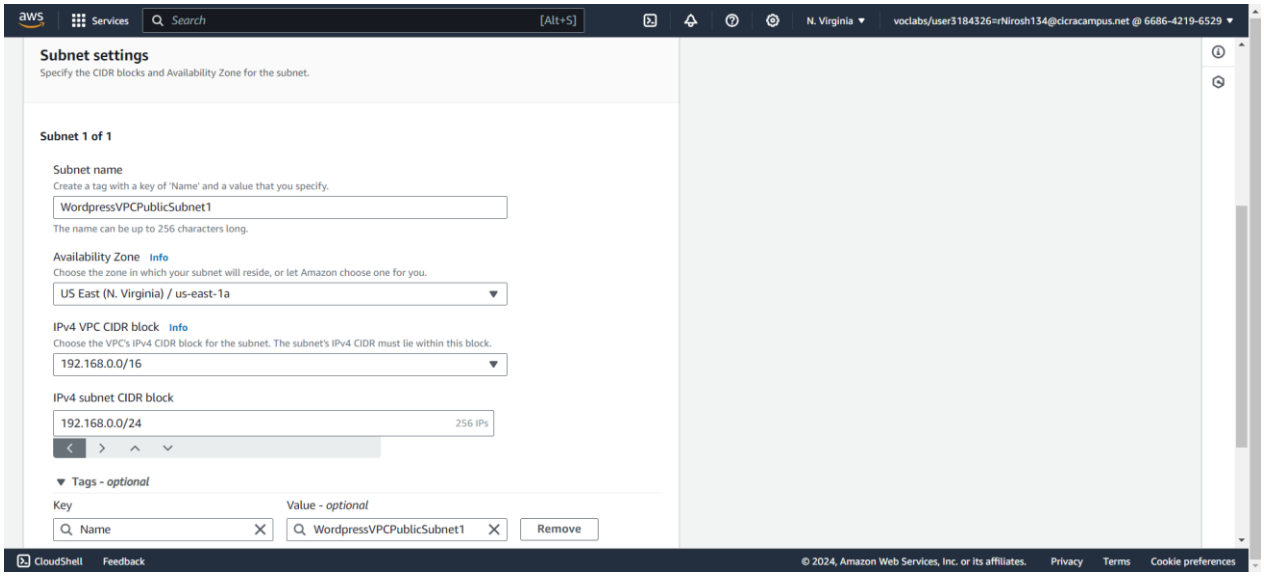
A. Create VPC

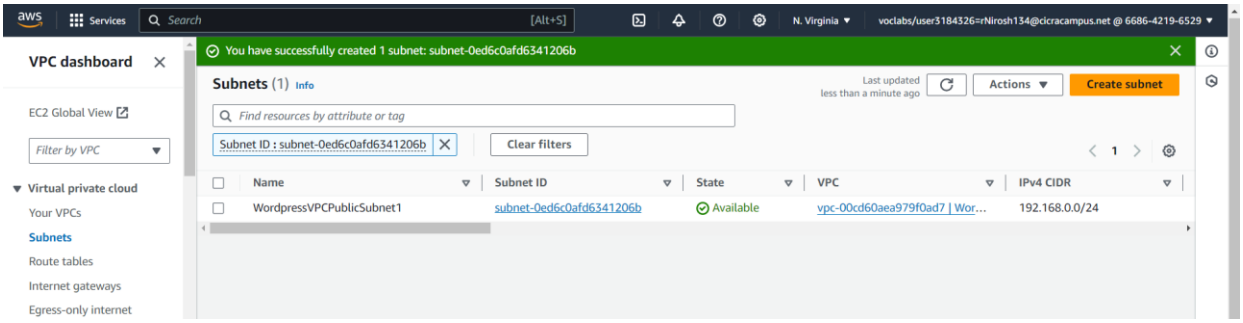
I started by navigating to the AWS Management Console's VPC Dashboard. I choose the VPC option from the Services menu in order to open the VPC Dashboard. I choose "Create VPC" after clicking on "Your VPCs" on the left navigation pane to start a new VPC. I completed the following fields on the VPC creation form: I left the tenancy as "Default," gave the VPC the name "WordPressVPC," and changed the IPv4 CIDR block to "10.0.0.0/16." At last, I selected "Create VPC" to finish the creation procedure.

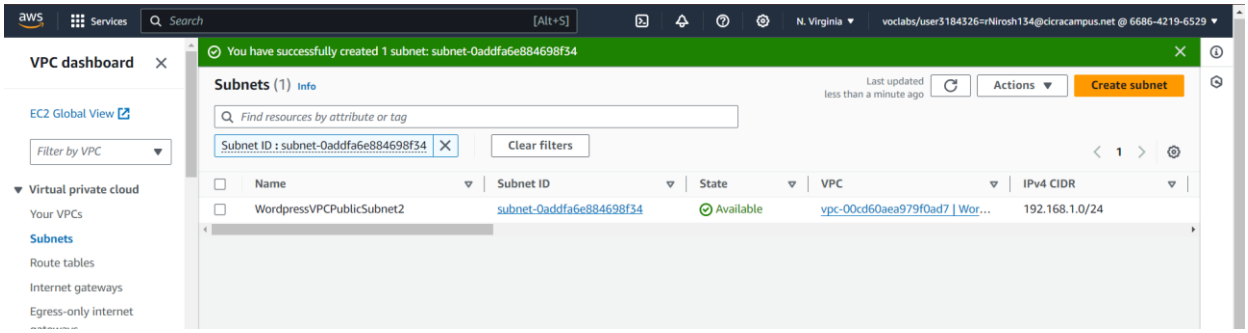


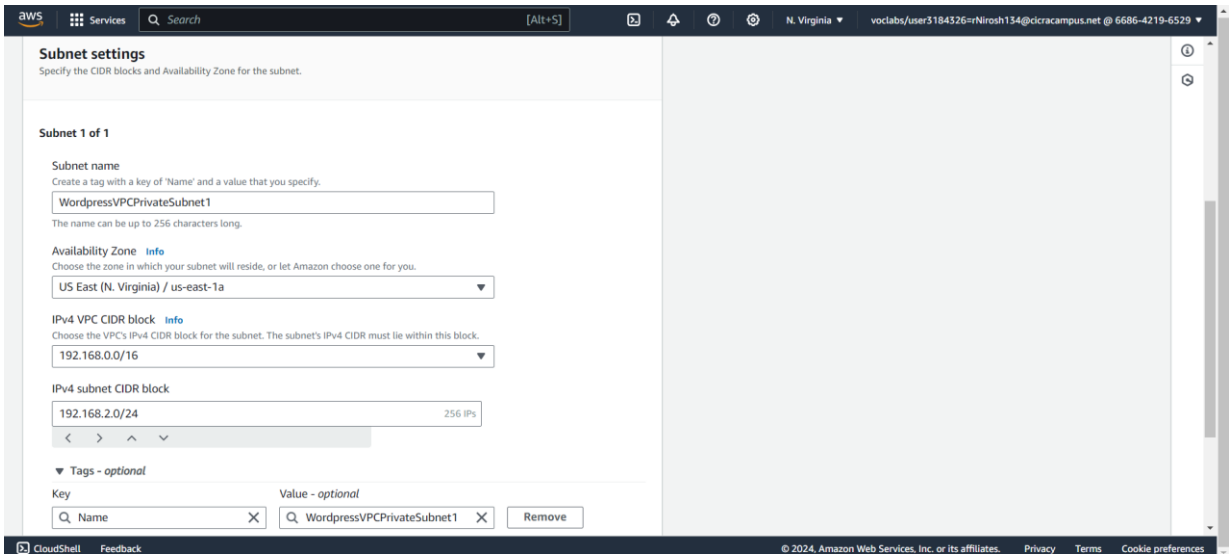
B. Create Subnets

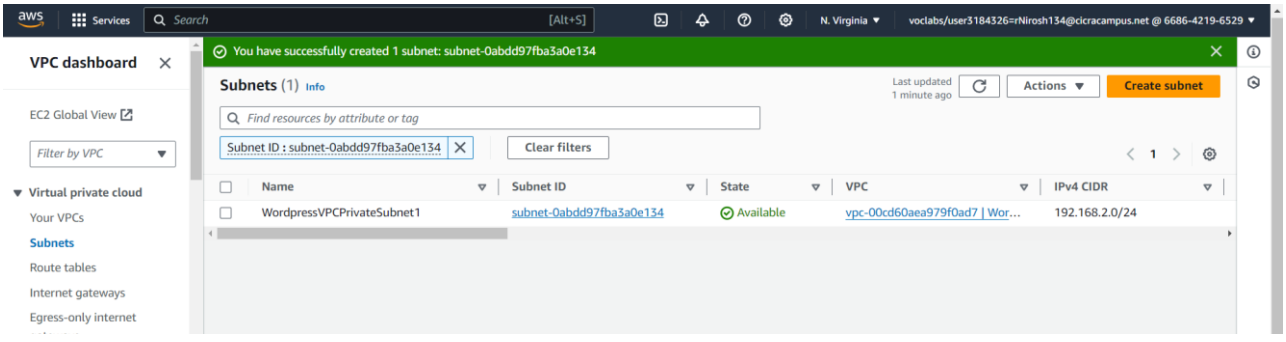
I began by selecting "Subnets" from the left navigation pane in order to configure the subnets. I chose "Create subnet" and filled in the following information to create the first public subnet: I gave it the name "PublicSubnet1", selected "WordPressVPC" as the virtual private cloud (VPC), "us-east-1a" as the availability zone, and "10.0.0.0/24" as the IPv4 CIDR block. The second public subnet was created using the same procedure; it was given the name "PublicSubnet2", a different availability zone (such as "us-east-1b"), and a CIDR block of "10.0.1.0/24". I took comparable actions for the private subnets. With "10.0.2.0/24" in "us-east-1a" and "10.0.3.0/24" in "us-east-1b," I established "PrivateSubnet1" and "PrivateSubnet2".

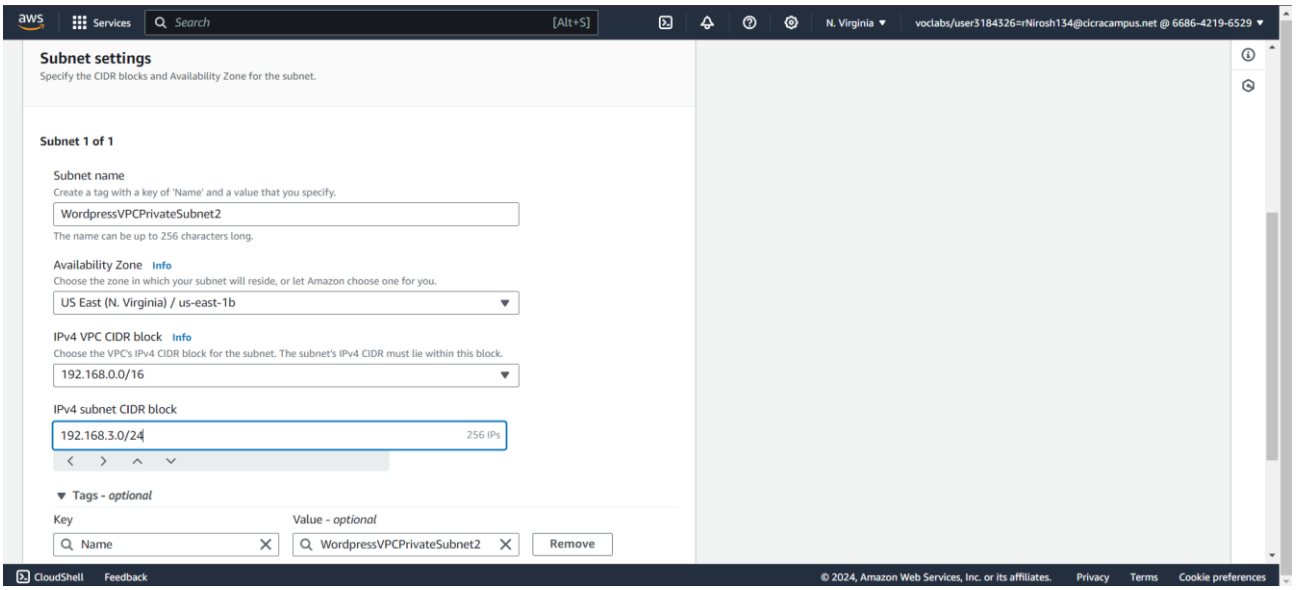


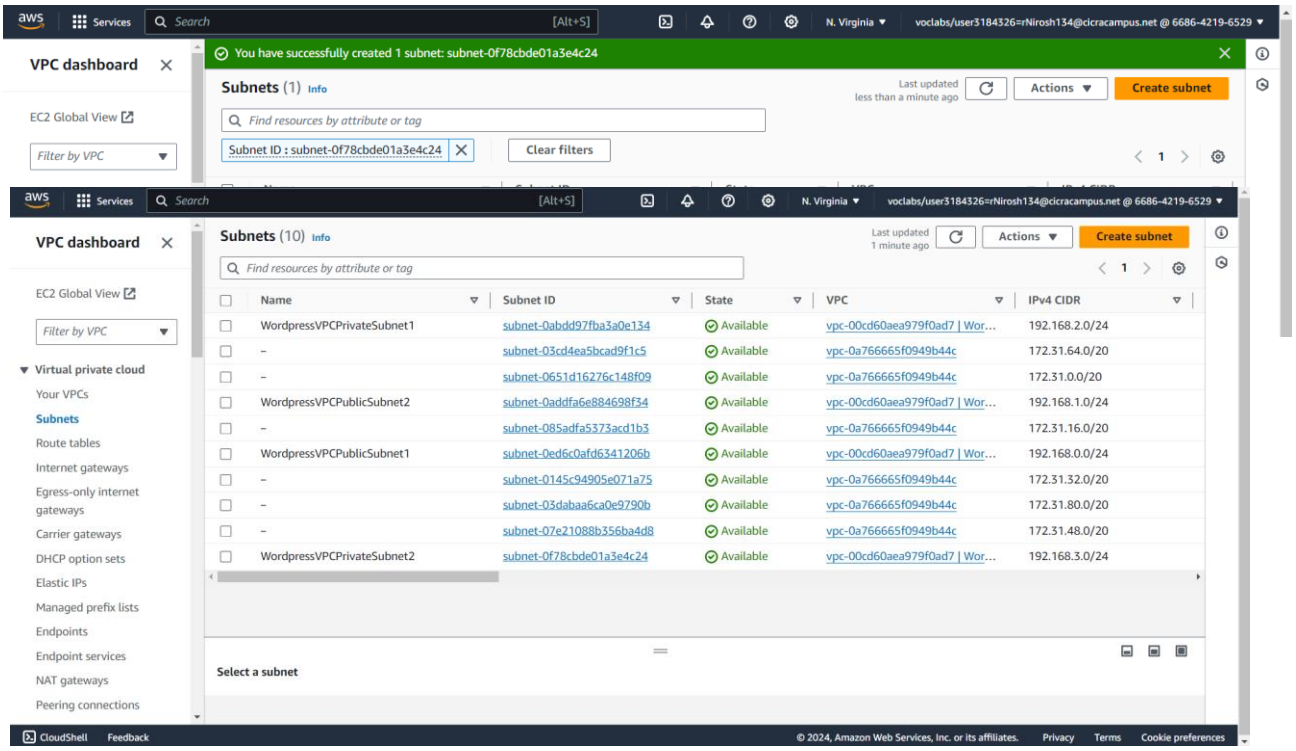

aws
Services
Search
[Alt+S]
N. Virginia
voclabs/user3184326=rNirosh134@cicracampus.net @ 6686-4219-6529
VPC dashboard
You have successfully created 1 subnet: subnet-0ed6c0afd6341206b
Subnets (1) Info
Last updated less than a minute ago
Actions
Create subnet
Find resources by attribute or tag
Subnet ID : subnet-0ed6c0afd6341206b
Clear filters
Name
Subnet ID
State
VPC
IPv4 CIDR
WordpressVPCPublicSubnet1
subnet-0ed6c0afd6341206b
Available
vpc-00cd60aea979f0ad7 | Wor...
192.168.0.0/24
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet


aws
Services
Search
[Alt+S]
N. Virginia
voclabs/user3184326=rNirosh134@cicracampus.net @ 6686-4219-6529
VPC dashboard
You have successfully created 1 subnet: subnet-0addfa6e884698f34
Subnets (1) Info
Last updated less than a minute ago
Actions
Create subnet
Find resources by attribute or tag
Subnet ID : subnet-0addfa6e884698f34
Clear filters
Name
Subnet ID
State
VPC
IPv4 CIDR
WordpressVPCPublicSubnet2
subnet-0addfa6e884698f34
Available
vpc-00cd60aea979f0ad7 | Wor...
192.168.1.0/24
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet


aws
Services
Search
[Alt+S]
N. Virginia
voclabs/user3184326=rNirosh134@cicracampus.net @ 6686-4219-6529
Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.
Subnet 1 of 1
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
WordpressVPCPrivateSubnet1
The name can be up to 256 characters long.
Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
US East (N. Virginia) / us-east-1a
IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/16
IPv4 subnet CIDR block
192.168.2.0/24
256 IPs
Tags - optional
Key
Value - optional
Name
WordpressVPCPrivateSubnet1
Remove
CloudShell
Feedback
© 2024, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


aws
Services
Search
[Alt+S]
N. Virginia
voclabs/user3184326=rNirosh134@cicracampus.net @ 6686-4219-6529
VPC dashboard
You have successfully created 1 subnet: subnet-0abdd97fba3a0e134
Subnets (1) Info
Last updated 1 minute ago
Actions
Create subnet
Find resources by attribute or tag
Subnet ID : subnet-0abdd97fba3a0e134
Clear filters
Name
Subnet ID
State
VPC
IPv4 CIDR
WordpressVPCPrivateSubnet1
subnet-0abdd97fba3a0e134
Available
vpc-00cd60aea979f0ad7 | Wor...
192.168.2.0/24
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet

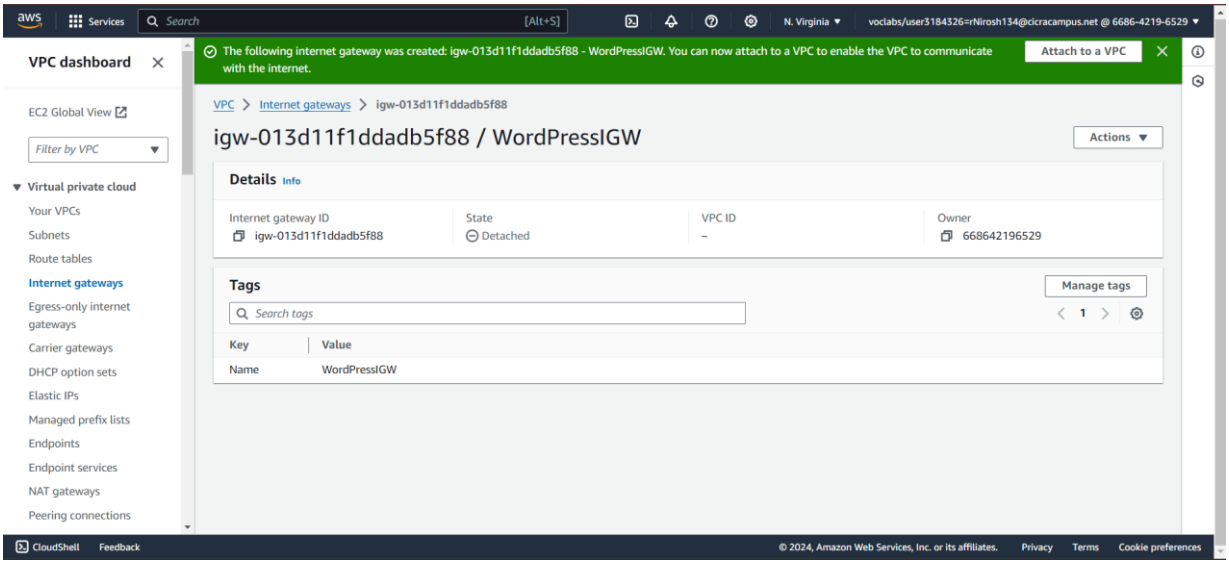


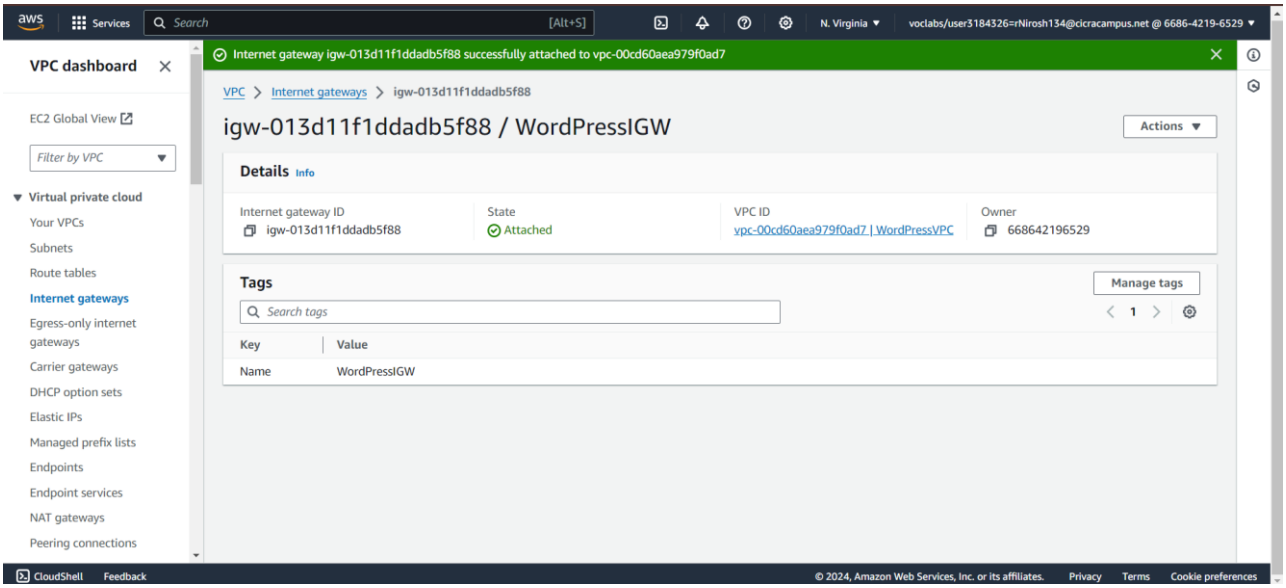


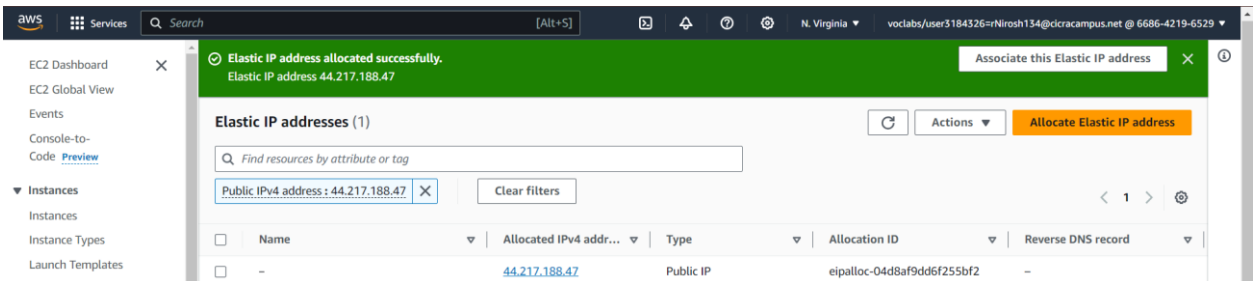
C. Create Internet Gateway

In order to facilitate communication between the VPC and the internet, I then developed an Internet Gateway. I selected "Create internet gateway" after selecting "Internet Gateways" in the left navigation pane. I gave it the name "WordPressIGW" and continued by selecting "Create internet gateway". I chose "WordPressIGW,"

clicked "Actions," selected "Attach to VPC," choose "WordPressVPC," and then I clicked "Attach internet gateway" to confirm that I wanted to attach this internet gateway to the VPC.

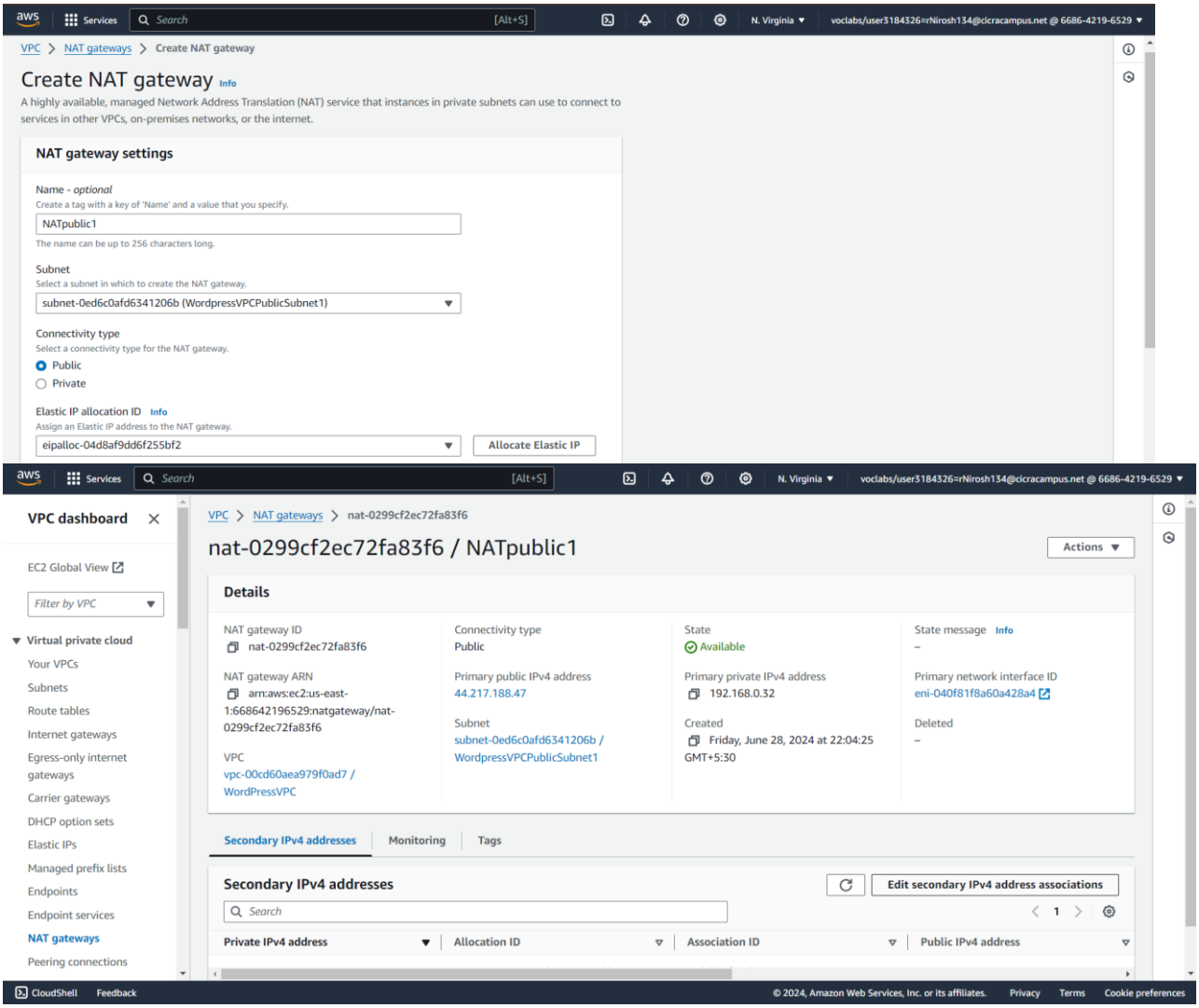






D. Create NAT Gateway

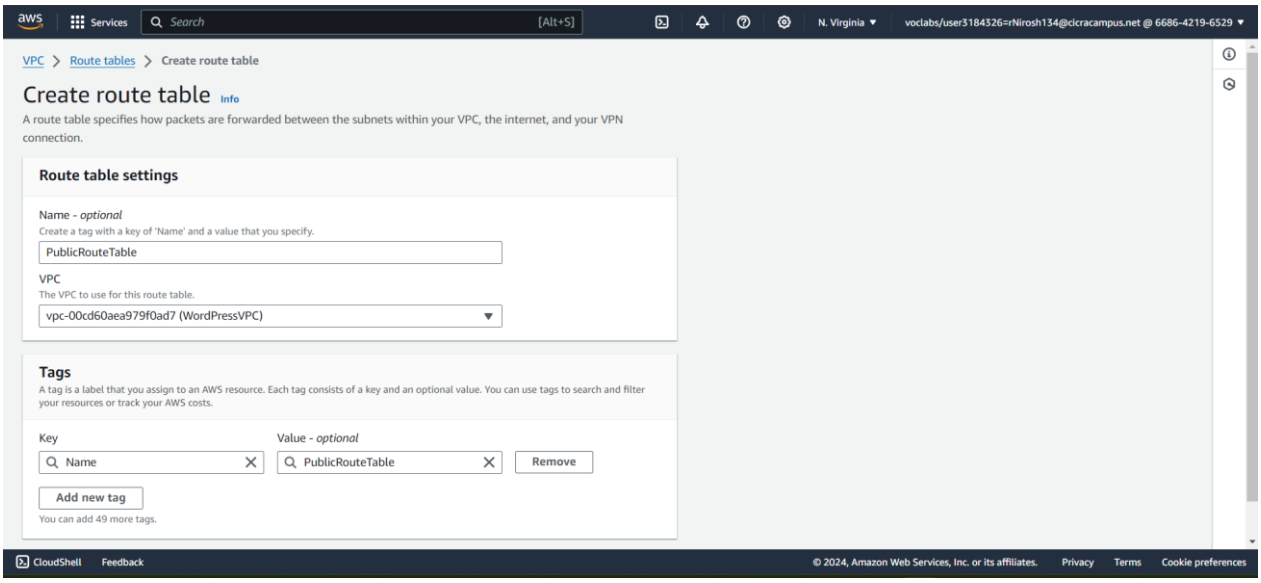
To access the internet through my private subnets, I required a NAT Gateway. Making an Elastic IP was my first step. I selected "Allocate Elastic IP address" and then "Allocate" after going to the "Elastic IPs" section of the EC2 Dashboard. I went back to the VPC Dashboard after setting up the Elastic IP, clicked on "NAT Gateways" on the left navigation pane, and chose "Create NAT Gateway". I picked the newly established Elastic IP, PublicSubnet1 as the subnet, and clicked "Create NAT Gateway".

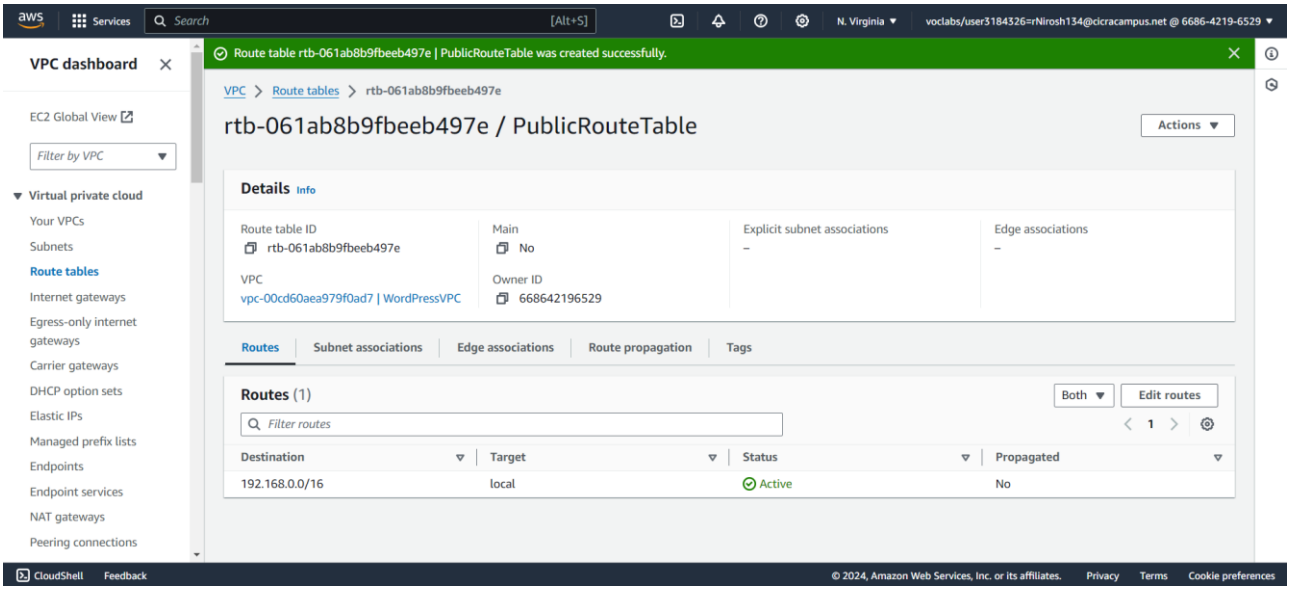


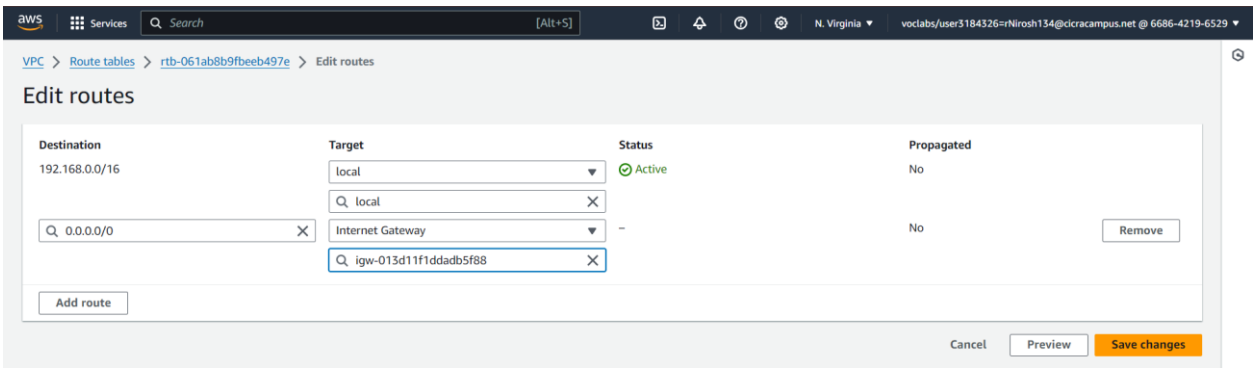
E. Create Route Tables

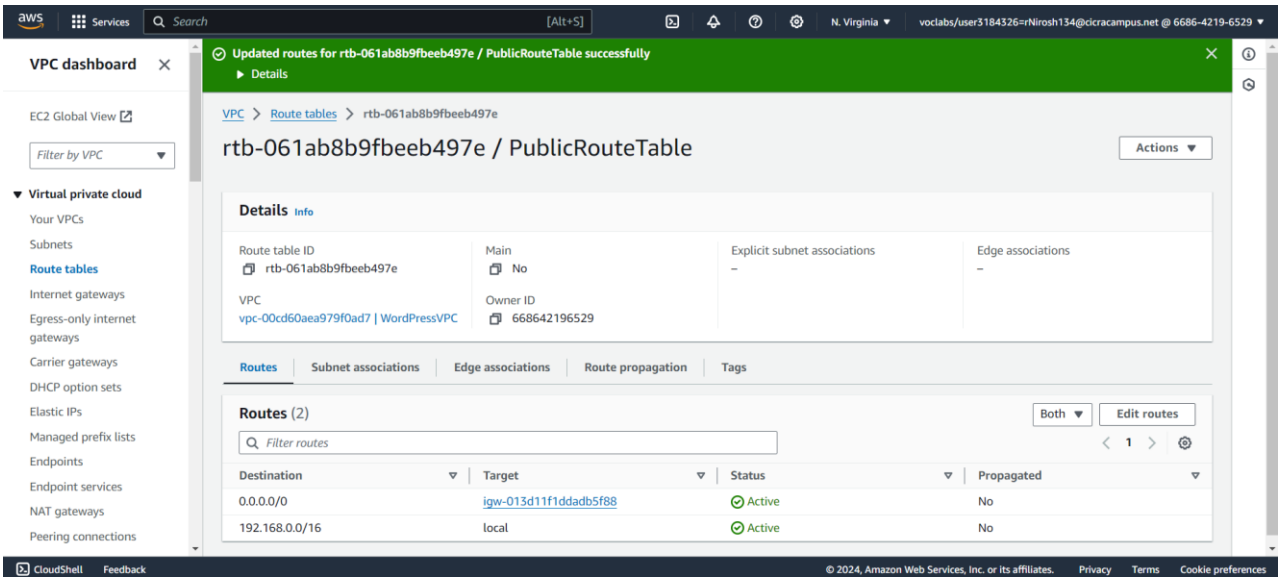
I developed a route table for the public subnets in order to control traffic routing within the VPC. I selected "Create route table" after going to "Route Tables" in the left navigation pane. With "PublicRouteTable" as its name, I chose "WordPressVPC" as the VPC. I edited the routes by choosing "PublicRouteTable," selecting the "Routes" tab, and then choosing "Edit routes" after generating the route table. I created a route and saved

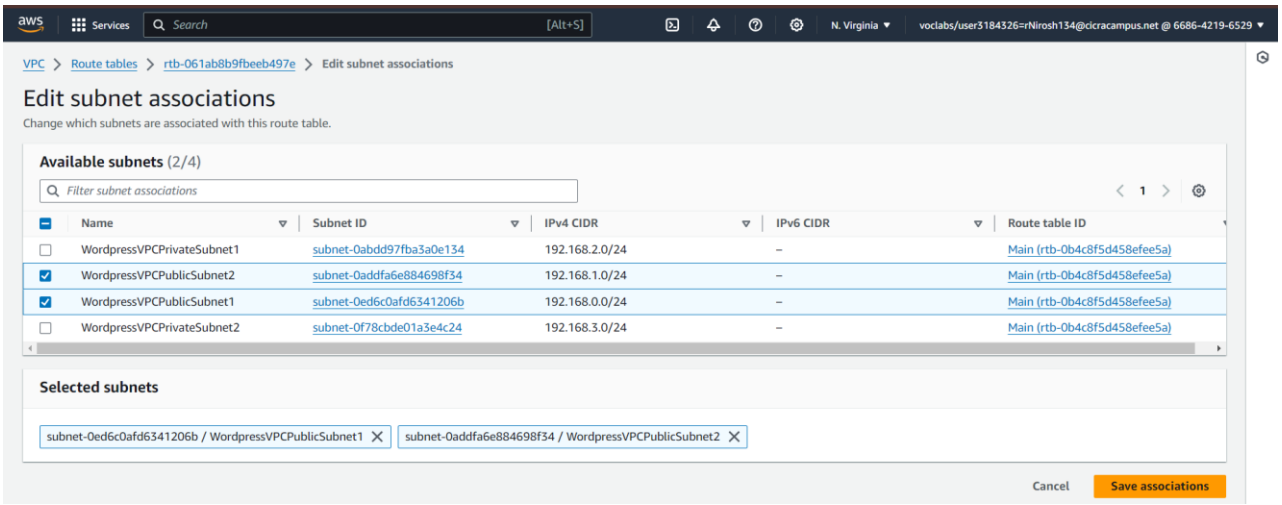
it with the destination set to "0.0.0.0/0" and the target set to "WordPressIGW". Lastly, I chose "PublicRouteTable," clicked on the "Subnet Associations" tab, chose "Edit subnet associations," picked "PublicSubnet1" and "PublicSubnet2," and saved the associations in order to associate the public subnets with this route table.

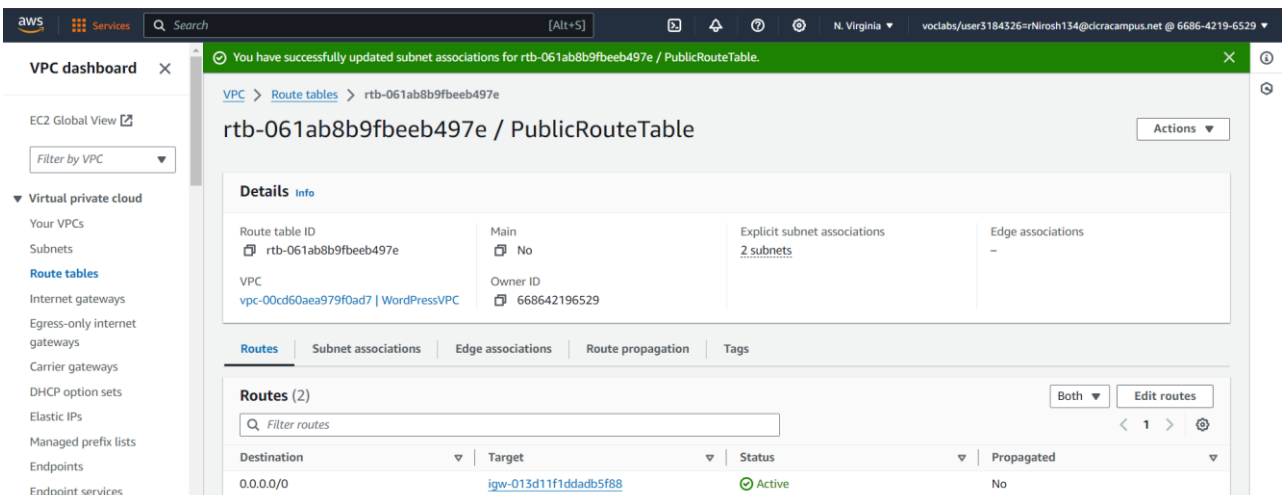


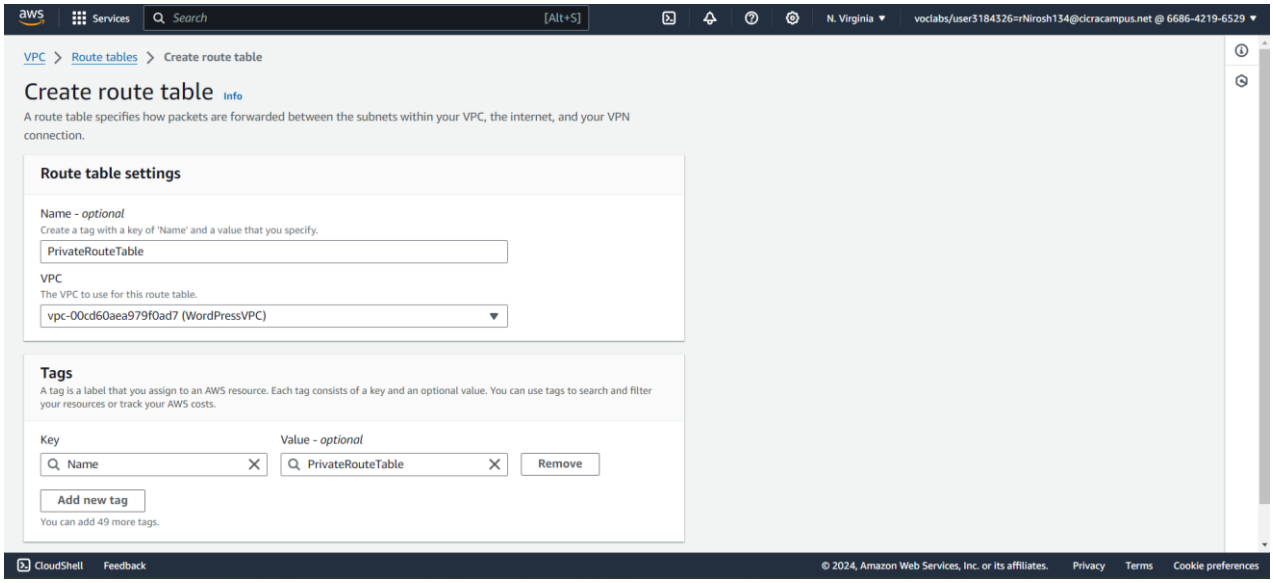


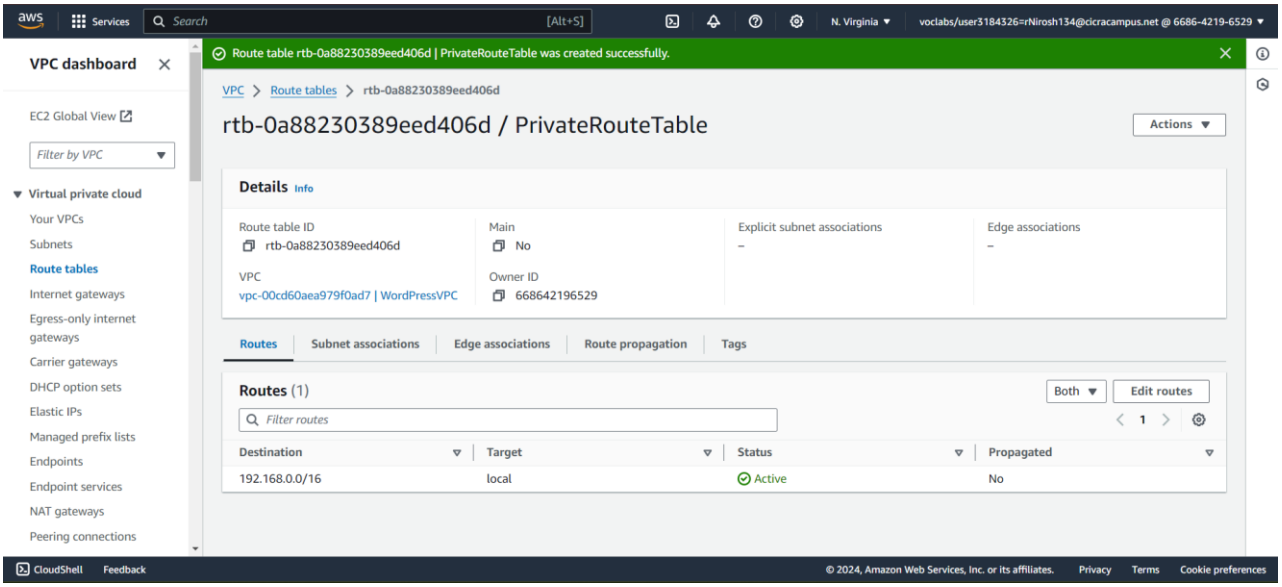


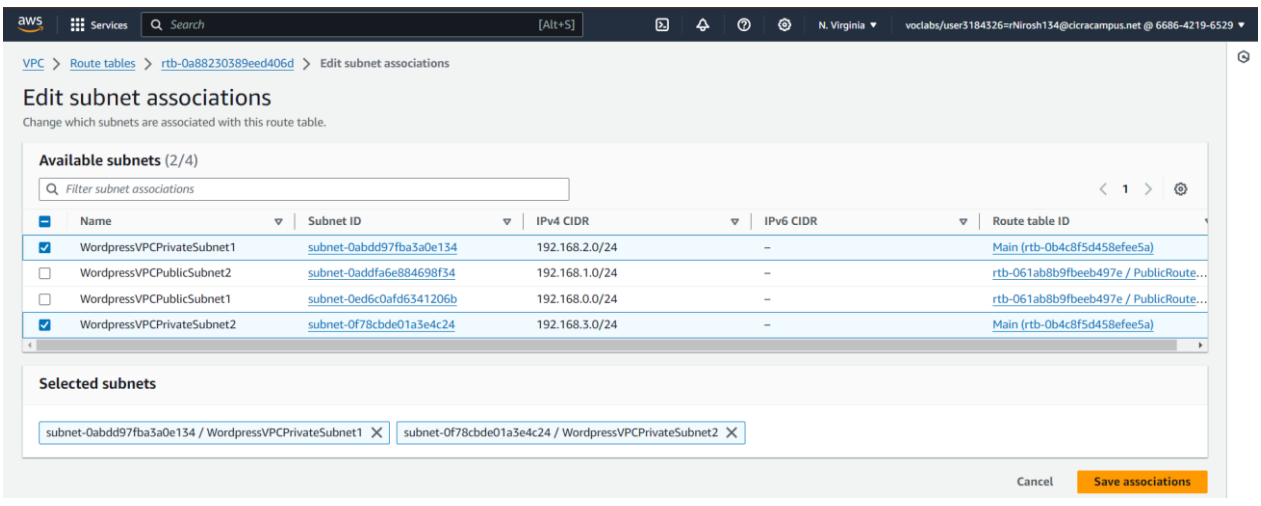

Updated routes for rtb-061ab8b9fbeeb497e / PublicRouteTable successfully
VPC dashboard
rtb-061ab8b9fbeeb497e / PublicRouteTable
Actions
Details
Routes (2)
Edit routes
Destination
Target
Status
Propagated
0.0.0.0/0
igw-013d11f1ddadb5f88
Active
No
192.168.0.0/16
local
Active
No

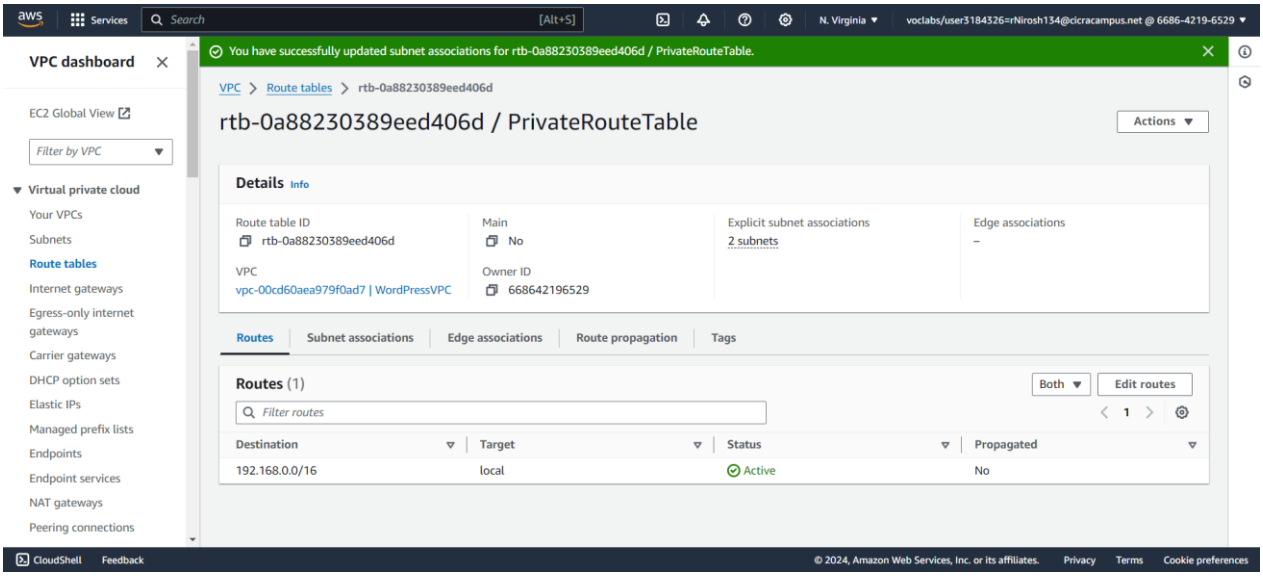

Edit subnet associations
Change which subnets are associated with this route table.
Available subnets (2/4)
WordpressVPCPrivateSubnet1
subnet-0abdd97fba3a0e134
192.168.2.0/24
WordpressVPCPublicSubnet2
subnet-0addfa6e884698f34
192.168.1.0/24
WordpressVPCPublicSubnet1
subnet-0ed6c0afd6341206b
192.168.0.0/24
WordpressVPCPrivateSubnet2
subnet-0f78cbde01a3e4c24
192.168.3.0/24
Selected subnets
subnet-0ed6c0afd6341206b / WordpressVPCPublicSubnet1
subnet-0addfa6e884698f34 / WordpressVPCPublicSubnet2
Cancel
Save associations

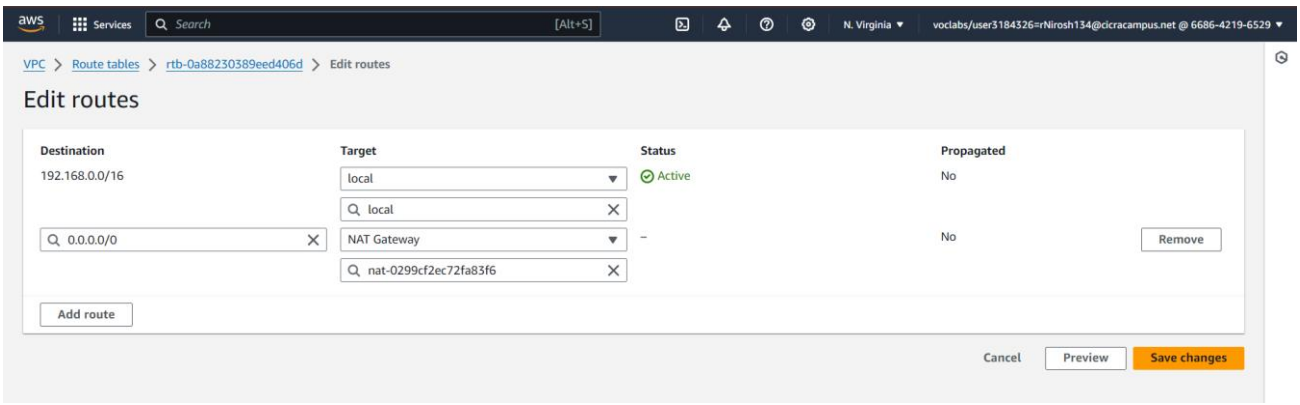

You have successfully updated subnet associations for rtb-061ab8b9fbeeb497e / PublicRouteTable.
VPC dashboard
rtb-061ab8b9fbeeb497e / PublicRouteTable
Actions
Details
Routes (2)
Edit routes
Destination
Target
Status
Propagated
0.0.0.0/0
igw-013d11f1ddadb5f88
Active
No

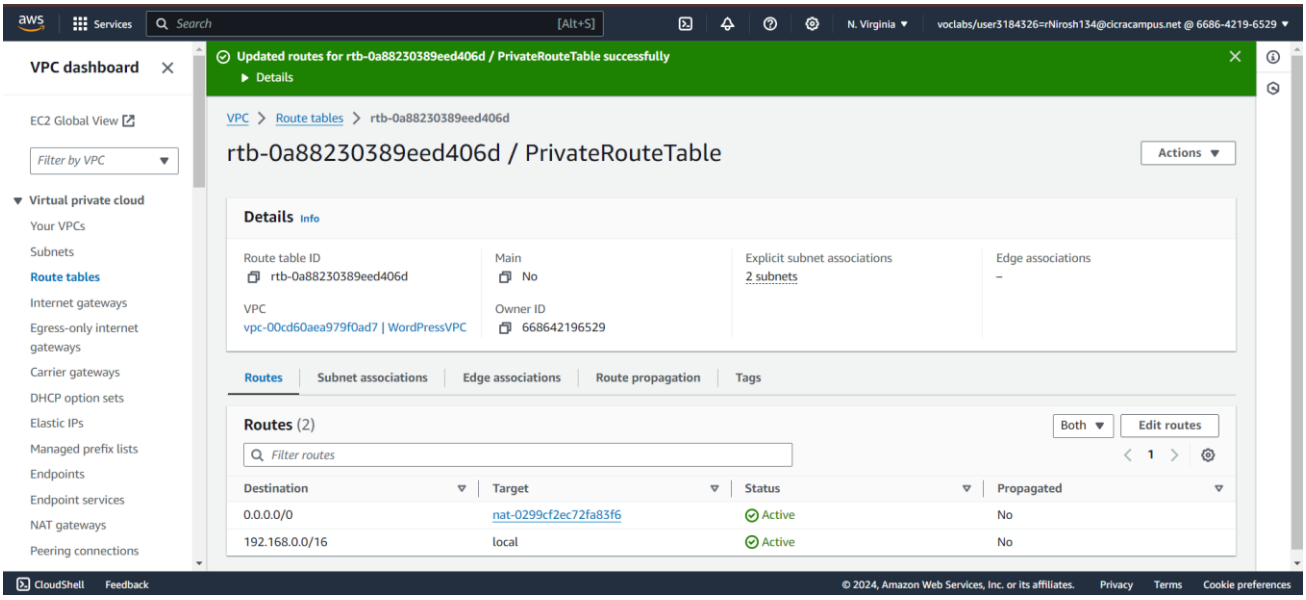

Services
Search
[Alt+S]
N. Virginia
VPC > Route tables > Create route table
Create route table
Info
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
PrivateRouteTable
VPC
The VPC to use for this route table.
vpc-00cd60aea979f0ad7 (WordPressVPC)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Value - optional
Name
PrivateRouteTable
Remove
Add new tag
You can add 49 more tags.
CloudShell
Feedback
© 2024, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Services
Search
[Alt+S]
N. Virginia
VPC dashboard
Route table rtb-0a88230389eed406d | PrivateRouteTable was created successfully.
VPC > Route tables > rtb-0a88230389eed406d
rtb-0a88230389eed406d / PrivateRouteTable
Actions
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
Endpoints
Endpoint services
NAT gateways
Peering connections
Details
Info
Route table ID
rtb-0a88230389eed406d
Main
No
Explicit subnet associations
–
Edge associations
–
VPC
vpc-00cd60aea979f0ad7 | WordPressVPC
Owner ID
668642196529
Routes
Subnet associations
Edge associations
Route propagation
Tags
Routes (1)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
192.168.0.0/16
local
Active
No
CloudShell
Feedback
© 2024, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Services
Search
[Alt+S]
N. Virginia
VPC > Route tables > rtb-0a88230389eed406d > Edit subnet associations
Edit subnet associations
Change which subnets are associated with this route table.
Available subnets (2/4)
Filter subnet associations
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
Route table ID
WordpressVPCPrivateSubnet1
subnet-0abdd97fba3a0e134
192.168.2.0/24
–
Main (rtb-0b4c8f5d458efee5a)
WordpressVPCPublicSubnet2
subnet-0addfa6e884698f34
192.168.1.0/24
–
rtb-061ab8b9fbeeb497e / PublicRoute...
WordpressVPCPublicSubnet1
subnet-0ed6c0afd6341206b
192.168.0.0/24
–
rtb-061ab8b9fbeeb497e / PublicRoute...
WordpressVPCPrivateSubnet2
subnet-0f78cbde01a3e4c24
192.168.3.0/24
–
Main (rtb-0b4c8f5d458efee5a)
Selected subnets
subnet-0abdd97fba3a0e134 / WordpressVPCPrivateSubnet1
subnet-0f78cbde01a3e4c24 / WordpressVPCPrivateSubnet2
Cancel
Save associations

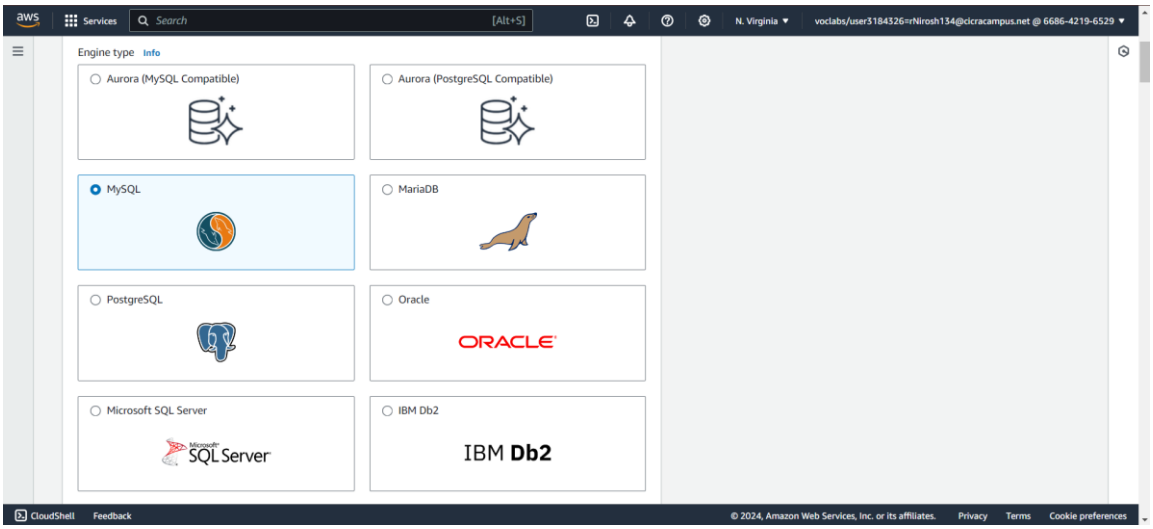

Services
Search
[Alt+S]
N. Virginia
voclabs/user3184326=rNirosh134@cicracampus.net @ 6686-4219-6529
VPC dashboard
You have successfully updated subnet associations for rtb-0a88230389eed406d / PrivateRouteTable.
VPC > Route tables > rtb-0a88230389eed406d
rtb-0a88230389eed406d / PrivateRouteTable
Actions
Details Info
Route table ID
rtb-0a88230389eed406d
Main
No
Explicit subnet associations
2 subnets
Edge associations
–
VPC
vpc-00cd60aea979f0ad7 | WordPressVPC
Owner ID
668642196529
Routes
Subnet associations
Edge associations
Route propagation
Tags
Routes (1)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
192.168.0.0/16
local
Active
No
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
Endpoints
Endpoint services
NAT gateways
Peering connections
CloudShell
Feedback
© 2024, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

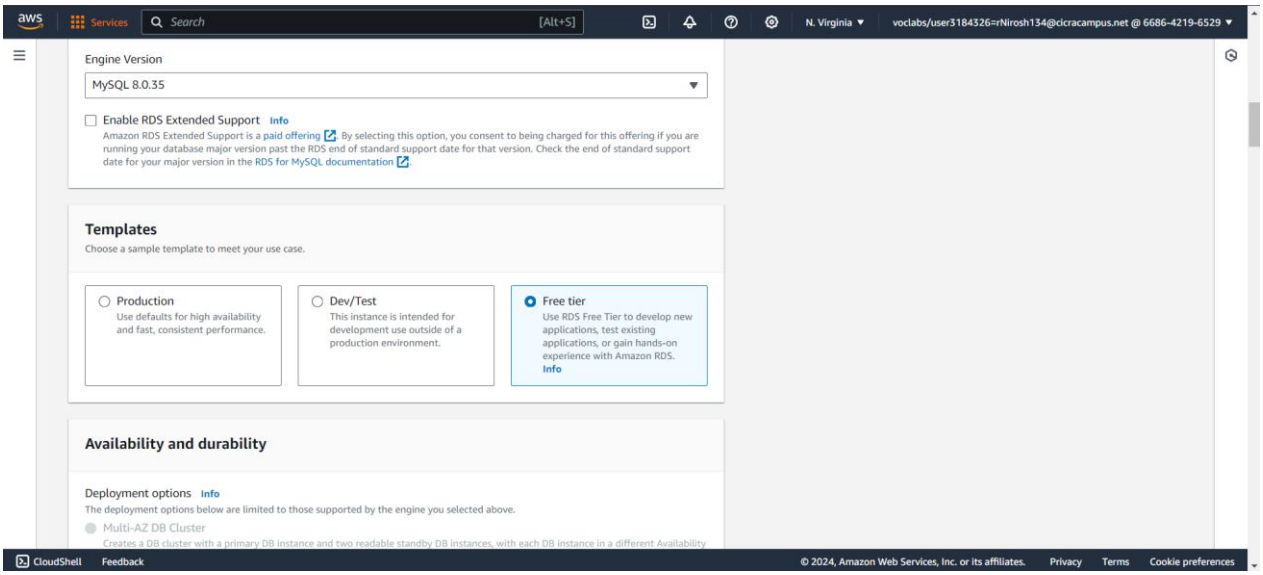

Services
Search
[Alt+S]
N. Virginia
voclabs/user3184326=rNirosh134@cicracampus.net @ 6686-4219-6529
VPC > Route tables > rtb-0a88230389eed406d > Edit routes
Edit routes
Destination
Target
Status
Propagated
192.168.0.0/16
local
Active
No
local
0.0.0.0/0
NAT Gateway
–
No
Remove
nat-0299cf2ec72fa83f6
Add route
Cancel
Preview
Save changes

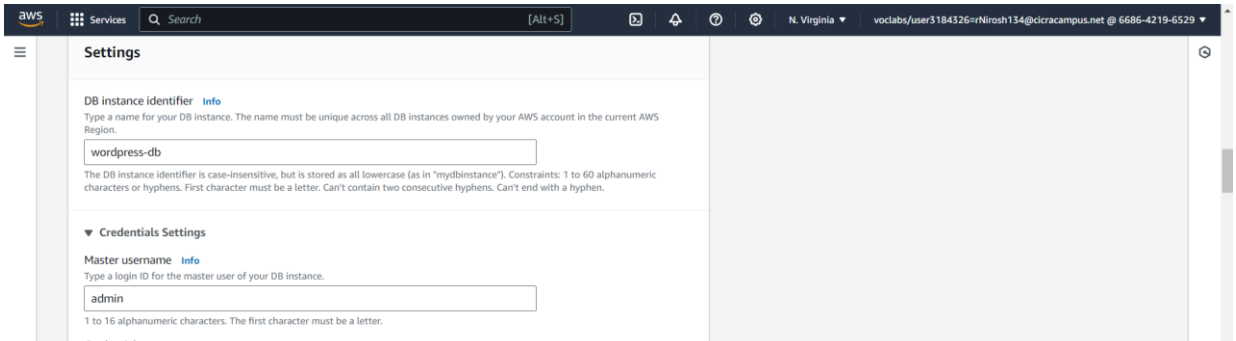

Services
Search
[Alt+S]
N. Virginia
voclabs/user3184326=rNirosh134@cicracampus.net @ 6686-4219-6529
VPC dashboard
Updated routes for rtb-0a88230389eed406d / PrivateRouteTable successfully
Details
VPC > Route tables > rtb-0a88230389eed406d
rtb-0a88230389eed406d / PrivateRouteTable
Actions
Details Info
Route table ID
rtb-0a88230389eed406d
Main
No
Explicit subnet associations
2 subnets
Edge associations
–
VPC
vpc-00cd60aea979f0ad7 | WordPressVPC
Owner ID
668642196529
Routes
Subnet associations
Edge associations
Route propagation
Tags
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
0.0.0.0/0
nat-0299cf2ec72fa83f6
Active
No
192.168.0.0/16
local
Active
No
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
Endpoints
Endpoint services
NAT gateways
Peering connections
CloudShell
Feedback
© 2024, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

F. Setup RDS

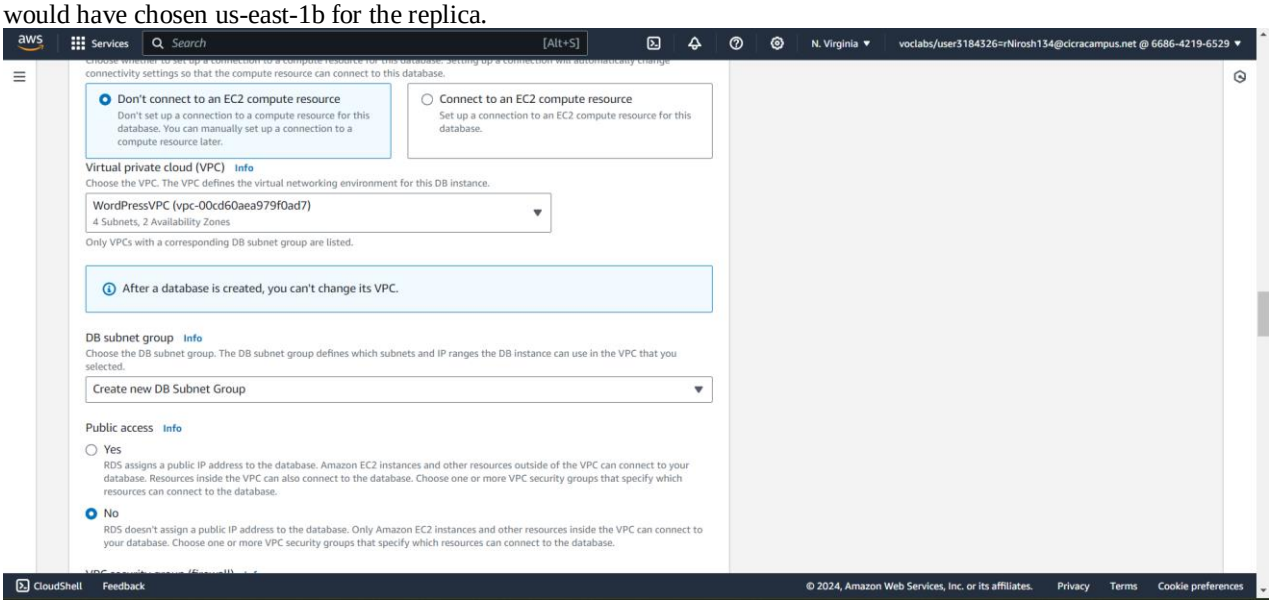
I had to set up a MySQL RDS instance with certain specifications in order to set up the database. After logging into the AWS Management Console, I used the Services menu to see the RDS Dashboard. I selected "Create database" after selecting "Databases" from the left navigation bar. I went with the free tier template, MySQL, and the regular create approach. I gave the instance of the database the name wordpress-db, set the master username to admin, and entered a strong password while configuring the database. I chose the instance class to be db.t3.micro and set aside 20 GB of general purpose (SSD) storage. I used PrivateSubnet1 and PrivateSubnet2 to build a new DB Subnet Group and put WordPressVPC as the VPC in the advanced settings.

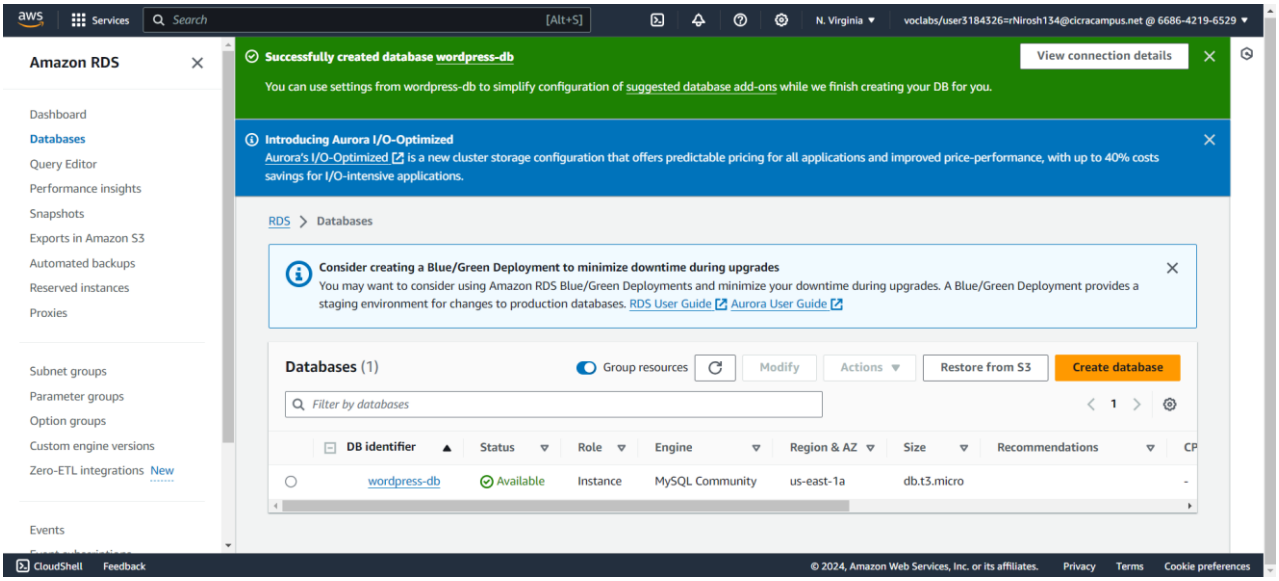


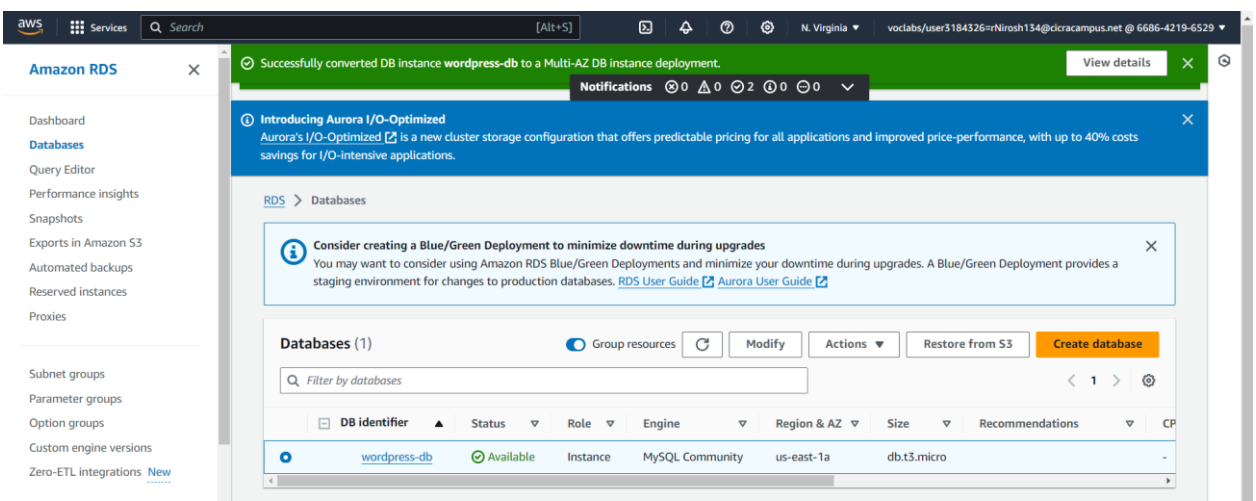


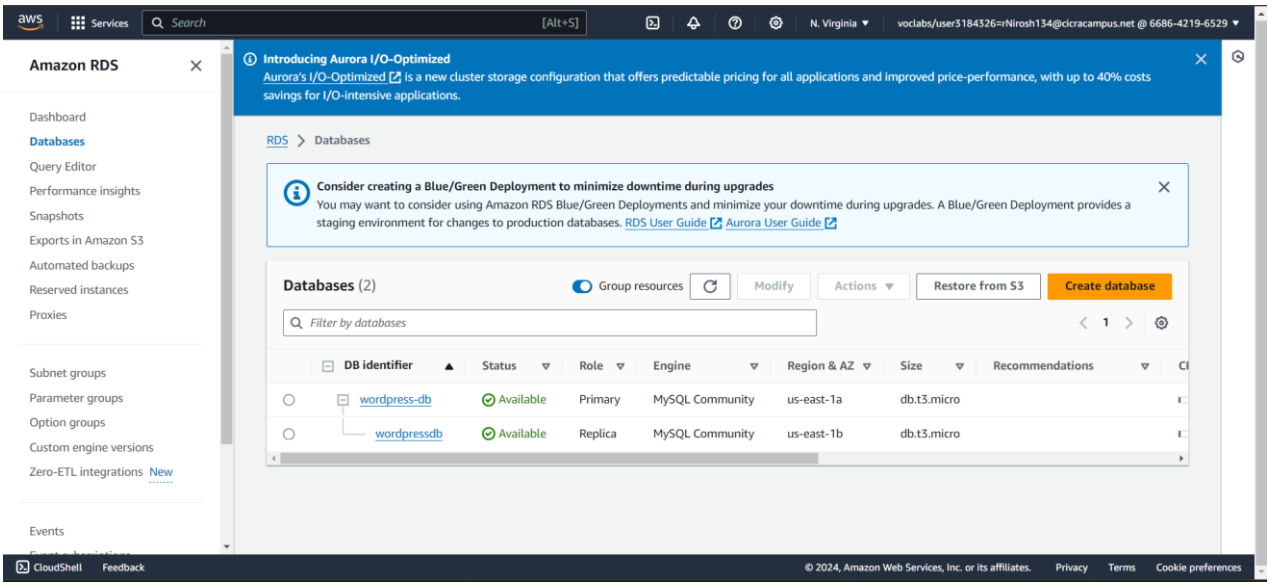


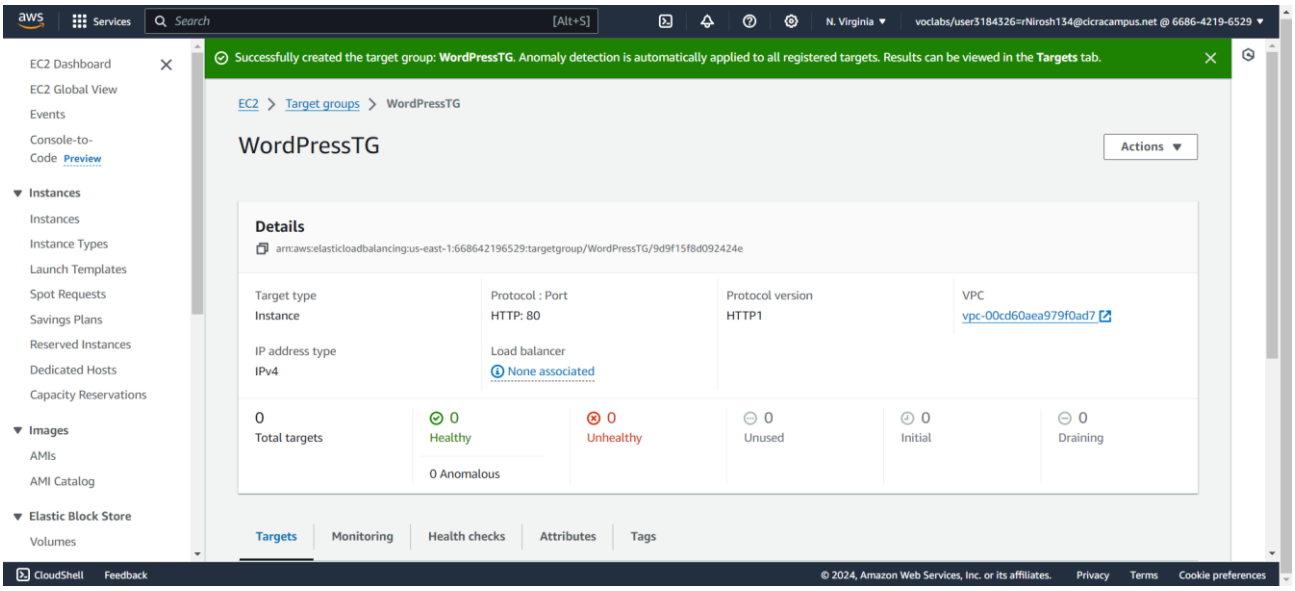
I made sure that all public access was blocked and made the wordpress-rds-sg security group, which permits MySQL/Aurora traffic on port 3306. I activated automated backups and entered WordPress as the database name. To increase the database's availability, I turned on Multi-AZ deployment under the availability and durability area. After the database was established, I picked the wordpress-db instance, clicked "Actions," and then selected "Create read replica." I selected db.t3.micro as the instance class and put the read replica in a different Availability Zone than the primary instance—for example, if the primary was in us-east-1a, I would have chosen us-east-1b for the replica.







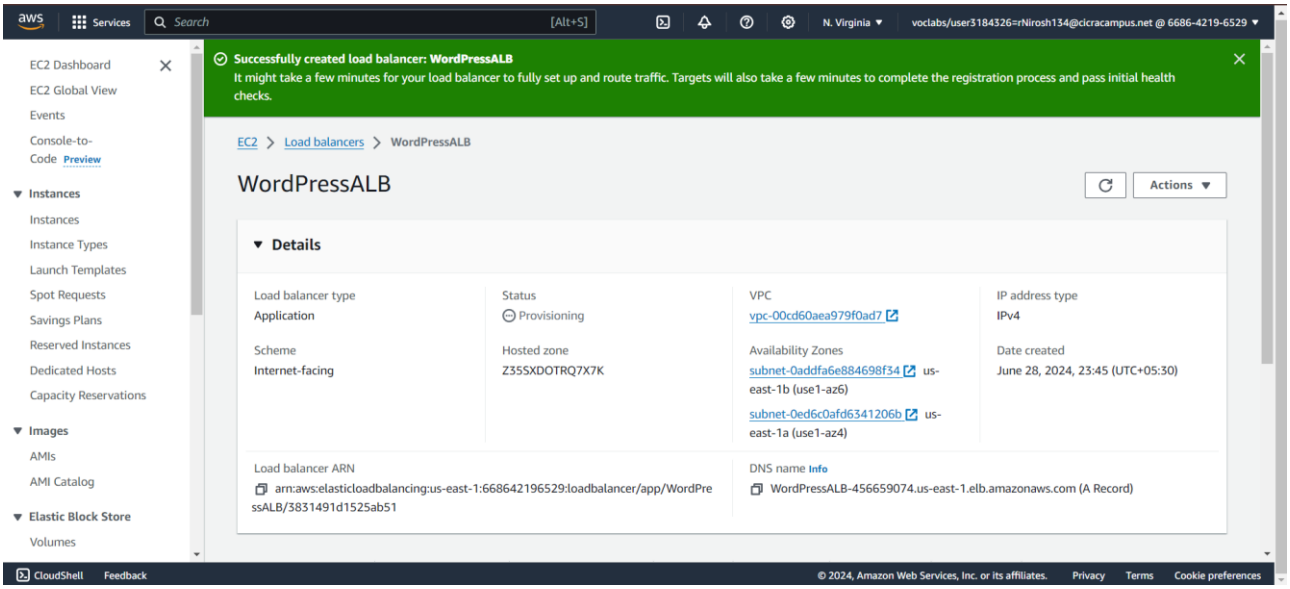




G. Configure ELB

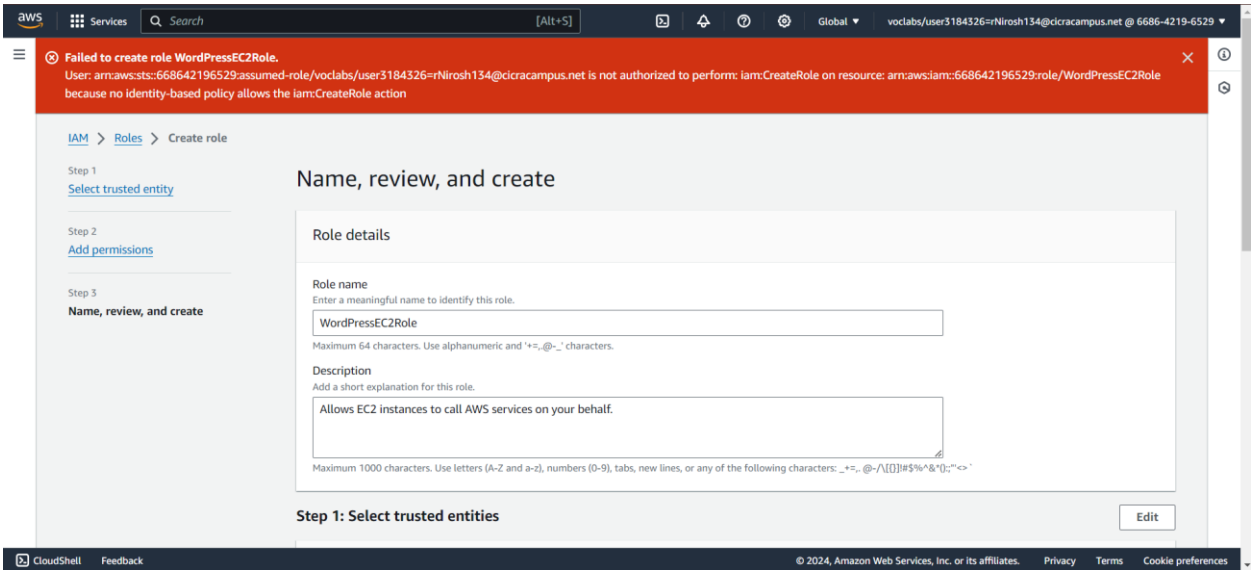
I configured an Application Load Balancer (ALB) to control the distribution of traffic among my web servers. I went to the EC2 Dashboard after logging into the AWS Management Console. I selected "Load Balancers" from the left navigation pane, followed by "Create Load Balancer". I chose the "Application Load Balancer" option, gave it the name WordPressALB, and configured the scheme to use IPv4 addresses for internet access. The load balancer is set up to listen on port 80. I put the ALB in the WordPressVPC's PublicSubnet1 and PublicSubnet2.

I built the wordpress-alb-sg security group in order to permit HTTP traffic on port 80 from the internet. I chose Instance as the target type, set the protocol to HTTP on port 80, and made a new target group called WordPressTG for routing. I registered targets after specifying these parameters because I would be adding the EC2 instances later.



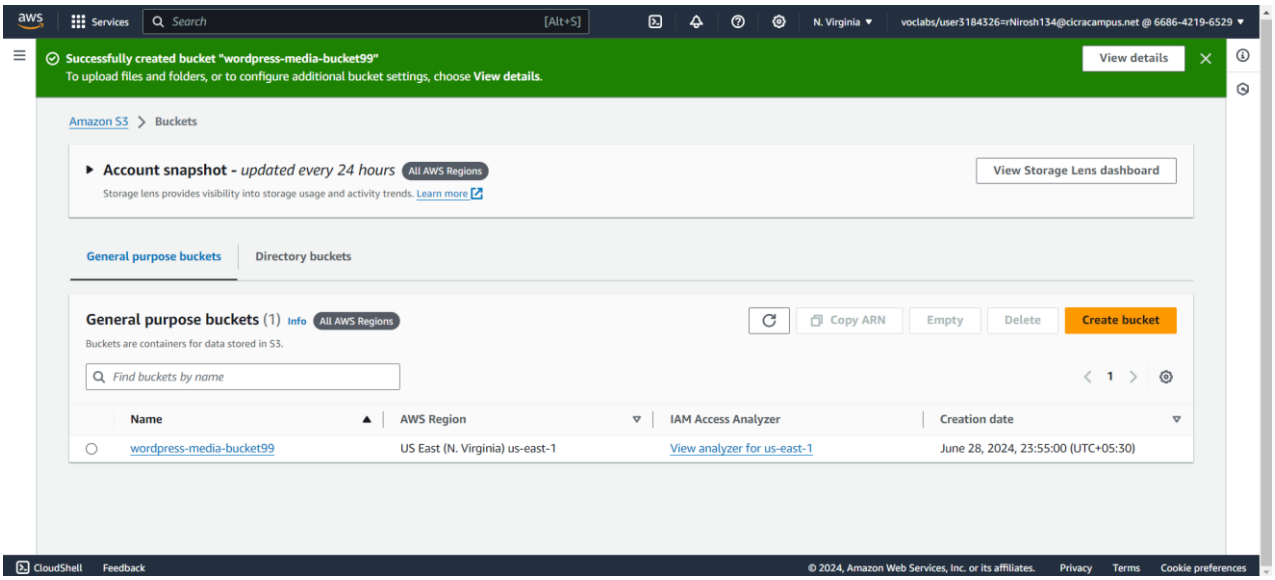
H. Create IAM Role

An error claiming that the IAM role could not be established appeared when I tried to create one. Because the creation of the IAM role is essential for allocating the required rights and roles to the resources in the VPC, this issue stopped me from moving forward with the project.



I. S3 Bucket Setup

I logged into the AWS Management Console and went to the Services menu to get the S3 Dashboard for the S3 bucket. I selected "Create bucket" and gave my bucket a catchy name, like wordpress-media-bucket. I choose the area I wanted and started the creation. I made sure that "Block all public access" was turned on, but I made a note that this option might not be available and that public access might still be enabled by default in the AWS Learner Labs environment.



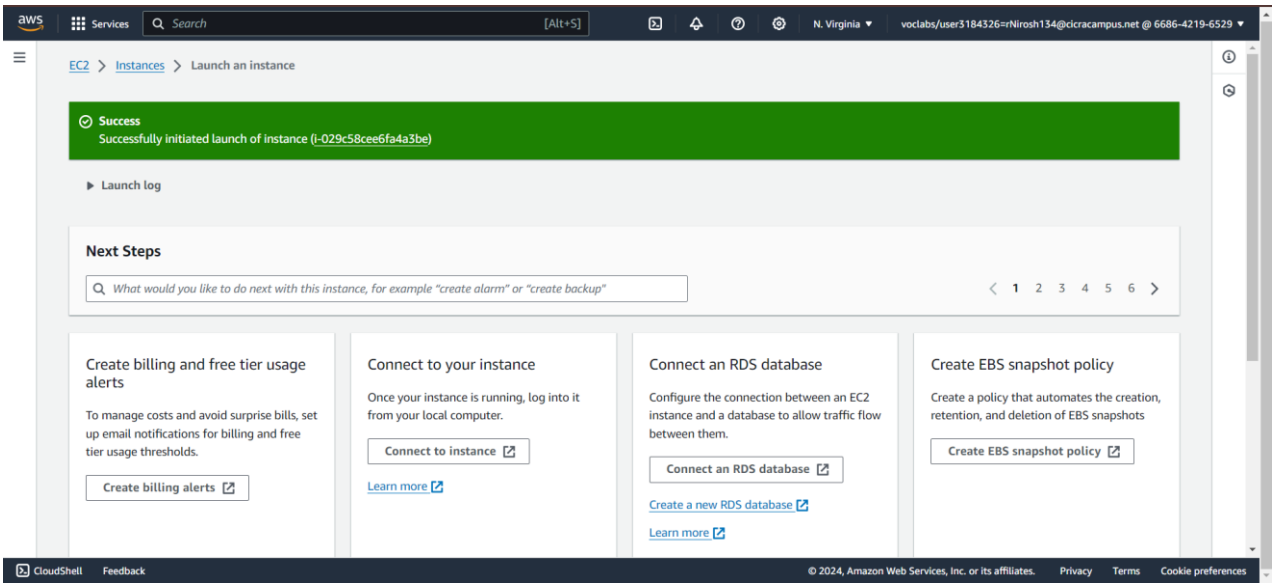
J. Launching EC2 Instance and Configuring WordPress

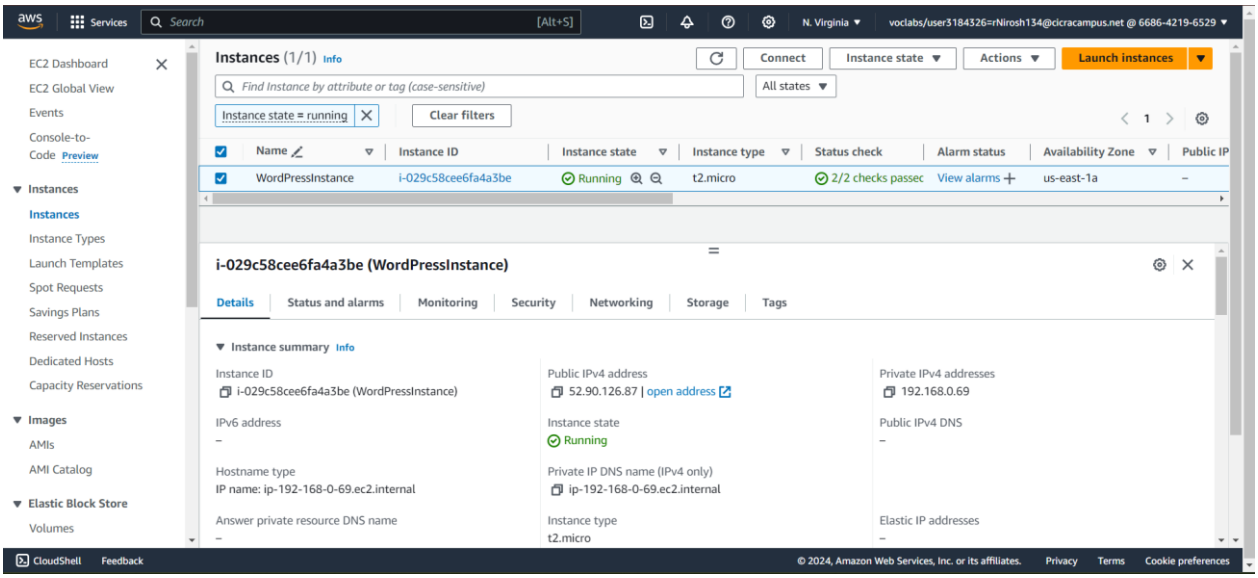
After that, I started an EC2 instance to host WordPress. I went to the EC2 Dashboard after logging into the AWS Management Console. I selected "Launch instances" after selecting "Instances" in the left navigation pane. I chose the Amazon Linux 2 AMI (HVM), SSD Volume Type for the Amazon Machine Image (AMI). Since the t2.micro instance type is available for free, that's why I went with it. For this EC2 instance, I made a fresh key pair so that I could access SSH later.

I then chose PublicSubnet1 as the subnet and set the instance to use the WordPressVPC. In order to make sure the instance could be accessed from the internet, I turned on Auto-assign Public IP. I opened up a new security group called wordpress-ec2-sg and allowed SSH (port 22) and HTTP (port 80). I tried attaching the WordPressEC2Role that was generated during the IAM setup within the IAM role section. Regretfully, I ran into a problem that prevented me from associating the IAM role here in the prior stage because it could not be created.

To configure the WordPress environment on the EC2 instance, I typed the following user data script in the Advanced Details section:

```
#!/bin/bash
yum -y update
yum -y install php httpd mysql
PHP_VERSION=`php -v | head -n 1 | awk '{print $2}' | awk -F "." '{print $1}'`
while [ ${PHP_VERSION} -ne 7 ]
do
amazon-linux-extras install php7.4 -y
PHP_VERSION=`php -v | head -n 1 | awk '{print $2}' | awk -F "." '{print $1}'`
done
yum -y install php-mbstring php-xml
wget http://wordpress.org/latest.tar.gz -P /tmp/
tar zxvf /tmp/latest.tar.gz -C /tmp
cp -r /tmp/wordpress/* /var/www/html/
chown apache:apache -R /var/www/html
systemctl enable httpd.service
systemctl start httpd.service
```





I had to wait a few minutes after the instance was launched before the status checks indicated "2/2" and the instance state changed to "running". I opened a browser and went to http://<EC2-Public-IP> after that. In order to configure the WordPress website, I would connect it to the RDS instance using the credentials I had already provided. This step should have brought up the WordPress initial setup screen.

But I was unable to finish the EC2 instance's entire setup since I could not create and attach the required IAM role. This stage was not completed because the security group settings and SSH access were not completely functional. As a result, I was unable to complete the project, which included installing WordPress settings on another EC2 instance in PublicSubnet2 and generating an AMI from the server.

IV. Discussion and Reflections

The project's goal was to use a variety of AWS services, such as VPC, subnets, RDS, ELB, S3, and EC2, to put up a WordPress website. The VPC, subnets, Internet gateway, NAT gateway, and route tables were successfully created in the first stages. In accordance with the project specifications, I was also able to set up the RDS instance, ELB, and S3 bucket. The IAM role creation procedure presented several obstacles for the project, nevertheless, and ultimately hindered the EC2 instance's successful launch and complete configuration. The inability to finish the WordPress setup and related configurations was caused by this problem. Notwithstanding these difficulties, the project offered insightful knowledge about the intricacies and dependencies of AWS services.

This configuration may improve the scalability, dependability, and security of online applications in practical business settings. The project's reflections highlight the significance of careful planning, role-based access control, and the requirement for troubleshooting abilities while utilizing cloud infrastructure to guarantee smooth application deployment and management.

V. **Conclusion**

The internet gateway, route tables, subnets, security groups, and VPC were all successfully created. However, a mistake in establishing the IAM position caused the project to be stopped at step 5, making further progress impossible. Steps 6 and 7 cannot be completed because they depend on the IAM role for security configurations and permissions. This was due to the inability to create the IAM role.

Acknowledgement

This report adheres to the NCWE 2018 conference's IEEE format template. We would especially like to appreciate the community forums and AWS documentation for their support during this endeavour.

My video link
https://www.youtube.com/watch?v=Aynk1SnPHSo